RQ Answers

omitted

2024-04-05

Data overview

```
glimpse(all_data)
```

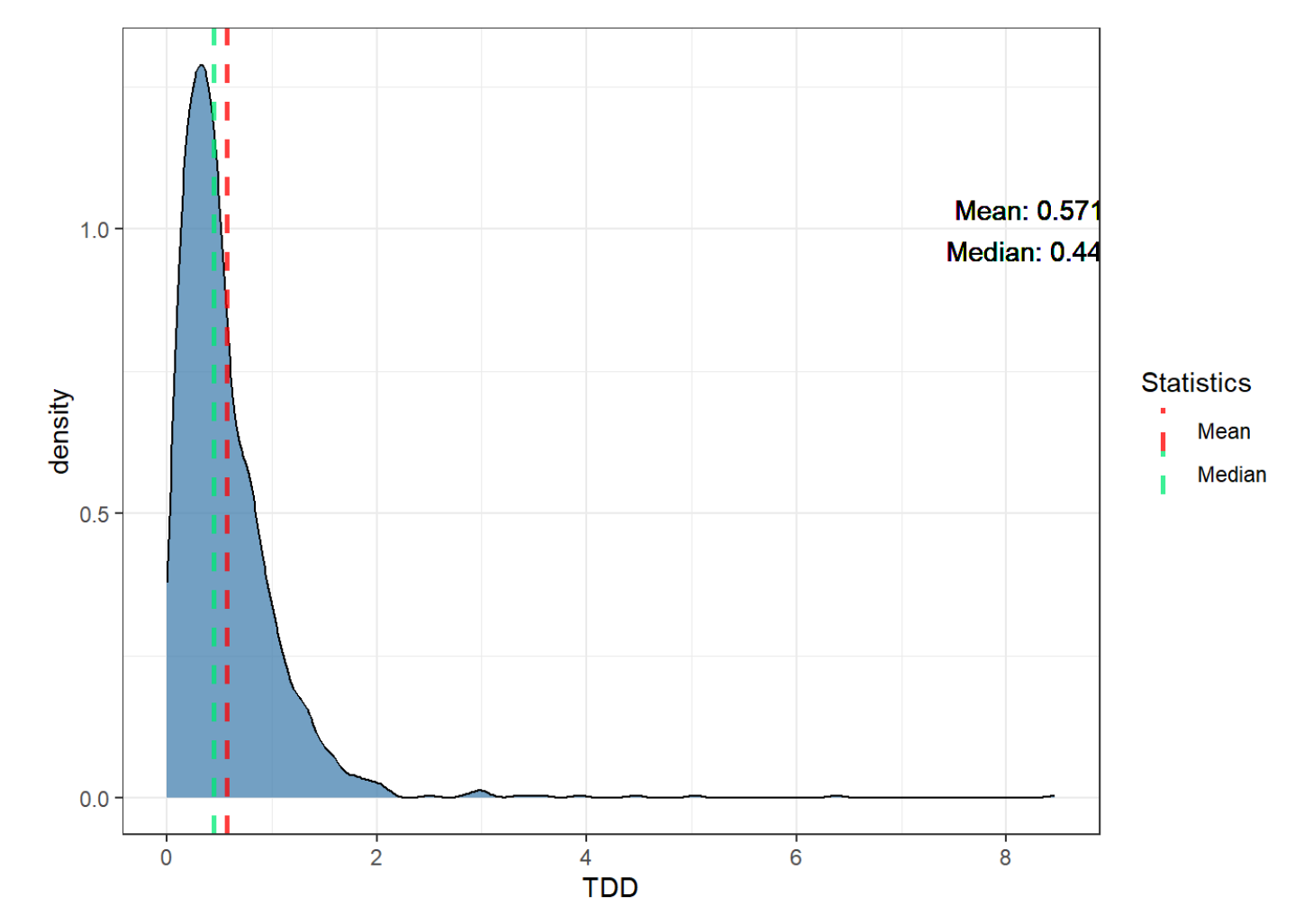
```
## Rows: 71,698
## Columns: 11
## $ repo                <chr> "accumulo", "accumulo", "accumulo", "accumulo", "a…
## $ pr_number           <int> 1302, 1302, 1302, 1302, 1302, 1302, 1302, 1302, 13…
## $ rule                <chr> "java:S5803", "java:S108", "java:S108", "java:S119…
## $ severity            <chr> "CRITICAL", "MAJOR", "MAJOR", "CRITICAL", "CRITICA…
## $ file                <chr> "server/tserver/src/test/java/org/apache/accumulo/…
## $ type                <chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM…
## $ status              <chr> "OPEN", "CLOSED", "CLOSED", "OPEN", "OPEN", "OPEN"…
## $ debt                <int> 10, 5, 5, 10, 10, 10, 10, 10, 20, 10, 10, 5, 5, 20…
## $ ncloc_affected_file <int> 96, 96, 96, 96, 96, 121, 121, 121, 249, 249, 249, …
## $ complexity          <int> 7, 7, 7, 7, 7, 6, 6, 6, 37, 37, 37, 37, 37, 37, 37…
## $ origin              <chr> "NEW", "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTI…
```

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 1152 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 1152 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

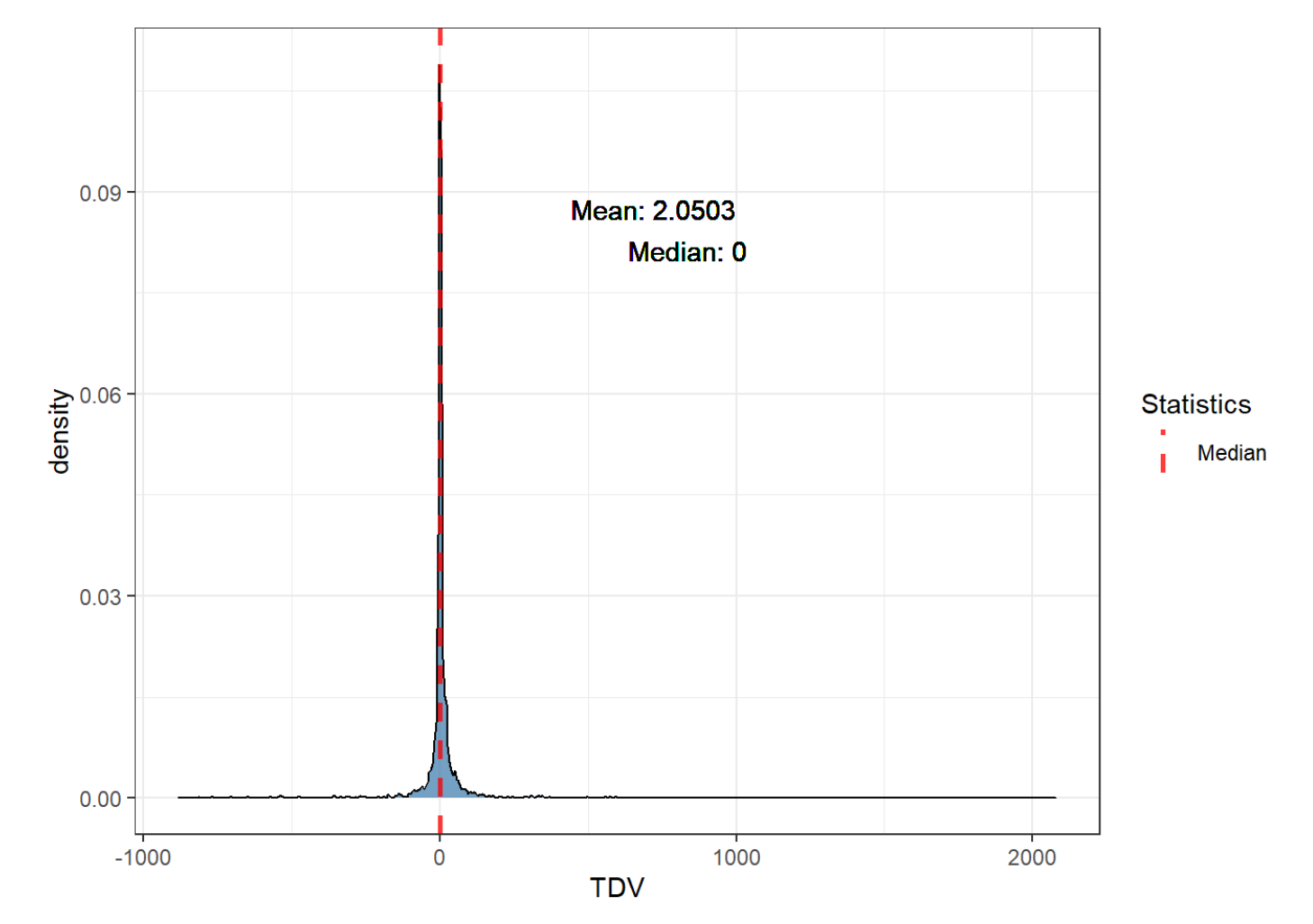


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 1152 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 1152 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 12 × 4
##    repo              tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##    <chr>                           <dbl>             <dbl>                 <dbl>
##  1 accumulo                        22.0               50.6                  27.3
##  2 cayenne                         11.1               40.7                  48.2
##  3 commons-collecti…               33.3               50                    16.7
##  4 commons-io                      28.8               50                    21.2
##  5 commons-lang                    21.8               63.6                  14.6
##  6 helix                           13.3               43.4                  43.4
##  7 httpcomponents-c…               16.7               43.8                  39.6
##  8 maven-surefire                  25                 65                    10
##  9 opennlp                         26.9               38.5                  34.6
## 10 struts                          26.7               36.7                  36.7
## 11 wicket                          12.5               50                    37.5
## 12 zookeeper                        9.09              54.6                  36.4
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 × 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##                      <dbl>                  <dbl>                      <dbl>
## 1                     20.6                   48.9                       30.5
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 12 × 3
##    repo                 preexisting_tdv_equal_to_zero preexisting_tdv_greater…¹
##    <chr>                                        <dbl>                     <dbl>
##  1 accumulo                                      62.6                      37.4
##  2 cayenne                                       68                        32
##  3 commons-collections                           50                        50
##  4 commons-io                                    66.7                      33.3
##  5 commons-lang                                  69.1                      30.9
##  6 helix                                         69.8                      30.2
##  7 httpcomponents-client                         64.4                      35.6
##  8 maven-surefire                                65                        35
##  9 opennlp                                       64.6                      35.4
## 10 struts                                        56.5                      43.5
## 11 wicket                                        61.5                      38.5
## 12 zookeeper                                     77.3                      22.7
## # ℹ abbreviated name: ¹preexisting_tdv_greater_than_zero
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##                           <dbl>                                  <dbl>
## 1                          64.6                                   35.4
```

Percentages for new TDV by repo

```
## # A tibble: 12 × 3
## # Groups:   repo [12]
##    repo                  new_tdv_equal_to_zero new_tdv_greater_than_zero
##    <chr>                                 <dbl>                     <dbl>
##  1 accumulo                               83.0                     17.0
##  2 cayenne                                93.3                      6.67
##  3 commons-collections                    80                       20
##  4 commons-io                             88.9                     11.1
##  5 commons-lang                           84.2                     15.8
##  6 helix                                  86.0                     14.0
##  7 httpcomponents-client                  95.6                      4.35
##  8 maven-surefire                        100                        0
##  9 opennlp                                96.2                      3.85
## 10 struts                                 97.9                      2.08
## 11 wicket                                 90                       10
## 12 zookeeper                              80                       20
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##                        <dbl>                          <dbl>
## 1                       89.6                           10.4
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (../issues_characterization.Rmd), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the $PositionScore$ metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^{k} Position_j$$

given a rule i, its $PositionScore_i$ will be the sum of its position across all k repositiries.After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 10 × 13
## # Groups:   rank [10]
##     rank accumulo      cayenne `commons-collections` `commons-io` `commons-lang`
##    <int> <chr>         <chr>   <chr>                 <chr>        <chr>
##  1     1 java:S1192  … java:S… java:S112  -  10      java:S4719 … java:S4165  -…
##  2     2 java:S112  -… java:S… java:S1170  -  6      java:S1192 … java:S1192  -…
##  3     3 java:S1135  … java:S… java:S3008  -  5      java:S899  … java:S125  - …
##  4     4 java:S100  -… java:S… java:S1874  -  5      java:S4042 … java:S2293  -…
##  5     5 java:S2095  … java:S… java:S125  -  4       java:S2129 … java:S3878  -…
##  6     6 java:S1155  … java:S… java:S1134  -  4      java:S1874 … java:S1612  -…
##  7     7 java:S1659  … java:S… java:S1192  -  3      java:S2293 … java:S100  - …
##  8     8 java:S2259  … java:S… java:S1126  -  2      java:S1481 … java:S1611  -…
##  9     9 java:S1854  … java:S… java:S1172  -  1      java:S125  … java:S4968  -…
## 10    10 java:S3776  … <NA>    java:S3011  -  1      java:S1764 … java:S3984  -…
## # ℹ 7 more variables: helix <chr>, `httpcomponents-client` <chr>,
## #   `maven-surefire` <chr>, opennlp <chr>, struts <chr>, wicket <chr>,
## #   zookeeper <chr>
```

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

```
## # A tibble: 58 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S1192             26
##  2 java:S112              70
##  3 java:S1874             74
##  4 java:S100              98
##  5 java:S125             106
##  6 java:S2293            112
##  7 java:S3740            115
##  8 java:S106             116
##  9 java:S1117            119
## 10 java:S1854            119
## # ℹ 48 more rows
```

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 × 13
## # Groups:   rank [10]
##     rank accumulo      cayenne `commons-collections` `commons-io` `commons-lang`
##    <int> <chr>         <chr>   <chr>                 <chr>        <chr>
##  1     1 java:S1192  … java:S… java:S1192  -  295    java:S1192 … java:S1192  -…
##  2     2 java:S112  -… java:S… java:S125  -  207     java:S100  … java:S100  - …
##  3     3 java:S1186  … java:S… java:S108  -  173     java:S899  … java:S4165  -…
##  4     4 java:S1659  … java:S… java:S1117  -  109    java:S4042 … java:S1172  -…
##  5     5 java:S2095  … java:S… java:S2293  -  78     java:S4719 … java:S125  - …
##  6     6 java:S106  -… java:S… java:S2201  -  46     java:S2184 … java:S3878  -…
##  7     7 java:S100  -… java:S… java:S3878  -  45     java:S117  … java:S108  - …
##  8     8 java:S3252  … java:S… java:S1854  -  36     java:S108  … java:S3400  -…
##  9     9 java:S2184  … java:S… java:S112  -  35      java:S3878 … java:S1104  -…
## 10    10 java:S108  -… java:S… java:S1123  -  33     java:S1874 … java:S1186  -…
## # ℹ 7 more variables: helix <chr>, `httpcomponents-client` <chr>,
## #   `maven-surefire` <chr>, opennlp <chr>, struts <chr>, wicket <chr>,
## #   zookeeper <chr>
```

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

```
## # A tibble: 51 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S1192             14
##  2 java:S100              86
##  3 java:S106              94
##  4 java:S112             103
##  5 java:S1186            103
##  6 java:S1874            106
##  7 java:S125             109
##  8 java:S2095            110
##  9 java:S108             112
## 10 java:S117             115
## # ℹ 41 more rows
```

Outliers

Higher TDD

```
## # A tibble: 1,152 × 5
##    repo           pr_number total_debt total_ncloc   tdd
##    <chr>              <int>      <int>       <int> <dbl>
##  1 wicket               709       2218         262  8.47
##  2 wicket               708       1161         182  6.38
##  3 struts               807        196          39  5.03
##  4 opennlp              563       1035         232  4.46
##  5 maven-surefire       596       1246         317  3.93
##  6 opennlp              569       1161         325  3.57
##  7 opennlp              585       1296         386  3.36
##  8 commons-lang        1051        978         324  3.02
##  9 helix               2439        102          34  3
## 10 struts               594        300         101  2.97
## # ℹ 1,142 more rows
```

Higher TDV

```
## # A tibble: 1,152 × 3
##    pr_number repo                    tdv
##        <int> <chr>                 <int>
##  1      2503 accumulo               2078
##  2      1891 accumulo                643
##  3       483 httpcomponents-client   593
##  4      1605 accumulo                574
##  5      2335 accumulo                557
##  6       559 struts                  496
##  7      2680 helix                   368
##  8      2289 accumulo                345
##  9      1452 helix                   343
## 10       427 struts                  333
## # ℹ 1,142 more rows
```

Lower TDV

```
## # A tibble: 1,152 × 3
##    pr_number repo                    tdv
##        <int> <chr>                 <int>
##  1       484 httpcomponents-client  -882
##  2       431 opennlp                -813
##  3      3604 accumulo               -768
##  4      1565 accumulo               -704
##  5       799 struts                 -648
##  6      3117 accumulo               -571
##  7       670 struts                 -542
##  8       826 struts                 -536
##  9       559 opennlp                -528
## 10      1584 helix                  -478
## # ℹ 1,142 more rows
```

Extra

Number of PRs with pre-existing TD

```
##      n
## 1 1094
```

Number of java:S1192 issues that affect test code

```
##       n
## 1 18504
```

Number of java:S1192 issues that affect not test code

```
##    n
## 1 84
```

Summary NCLOC

```
##      ncloc
##  Min.   :    3
##  1st Qu.:  180
##  Median :  414
##  Mean   : 1107
##  3rd Qu.: 1024
##  Max.   :61510
```